

National Road Complaint Management System

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Abstract—Road infrastructure is one of the most important public assets for safe transportation and economic development. However, problems such as potholes, broken roads, faulty streetlights, clogged drains, and missing traffic signs often remain unresolved because traditional reporting mechanisms are slow, manual, and lack transparency. Citizens frequently have limited visibility into the progress of their complaints, while local government workers face difficulties in organizing, prioritizing, and monitoring repair activities.

This paper presents a National Road Complaint Management System (RCMS), a digital platform that enables citizens to report road-related problems using images, descriptions, and geographic location information. The proposed system provides complaint tracking, administrator-based verification, worker task assignment, repair-status monitoring, and reporting facilities. The system integrates location-based reporting and centralized complaint management to improve communication between citizens and authorities. Analytical reports can further help identify frequently affected locations, monitor worker performance, estimate complaint resolution time, and support future road maintenance planning. The proposed approach aims to improve transparency, accountability, accessibility, and efficiency in road complaint management.

Index Terms—Road Complaint Management System, Smart City, GIS, GPS, E-Governance, Road Maintenance, Complaint Tracking, Web Application

I. INTRODUCTION

Road infrastructure is an important component of transportation, economic development, and public safety. Poor road conditions can affect daily transportation and may increase safety risks. Common road-related problems include potholes, damaged road surfaces, non-functional streetlights, blocked drainage systems, and missing traffic signs. Efficient monitoring and maintenance of road infrastructure are therefore important for maintaining safe and reliable transportation systems [1], [2].

Traditional methods of reporting road-related problems commonly depend on physical visits, telephone calls, or paper-based complaints. Such approaches can be time-consuming and may provide limited visibility into the progress of submitted complaints. Citizens may not know whether their complaints have been verified, assigned, or resolved. This lack of transparency can reduce confidence in the complaint-management process.

Modern digital technologies provide an opportunity to improve public complaint management. Web applications, geographic information systems (GIS), mobile reporting, and

image-based technologies can help collect and organize road-related information systematically [3]–[5].

The proposed Road Complaint Management System (RCMS) provides a centralized digital platform through which citizens can submit road complaints using photographs, descriptions, complaint categories, and geographic locations. The system enables administrators to verify complaints and assign them to maintenance workers. Workers can subsequently update the repair status and upload completion evidence.

The proposed system also provides analytical capabilities. Complaint records can be analyzed according to location, category, resolution time, and status. Such information can support infrastructure planning and help authorities identify frequently affected areas.

The major contributions of the proposed work are:

- Development of a centralized digital platform for reporting road-related complaints.
- GPS-based complaint registration with image upload.
- Real-time complaint tracking and notification mechanism.
- Administrator dashboard for complaint verification and task assignment.
- Worker module for updating repair progress and uploading completion evidence.
- Reporting and analytics module for hotspot identification and infrastructure planning.

The remainder of this paper is organized as follows. Section II presents the Literature Review. Section III discusses the Research Gap. Section IV explains the Proposed Methodology. Section V describes the System Architecture and Modules. Section VI presents the Results and Discussion. Finally, Section VII concludes the paper and discusses future work.

II. LITERATURE REVIEW

Several studies have investigated technologies and approaches for road maintenance, defect detection, citizen reporting, and public-service complaint management.

Image-processing techniques have been proposed for detecting potholes from asphalt pavement images. Such approaches can reduce the dependence on manual road inspection; however, their performance can depend on image quality and environmental conditions [1].

Studies of road maintenance systems in urban environments emphasize the importance of timely repair and effective infrastructure management. Poor integration between different

maintenance activities can remain a challenge for practical implementation [2].

Web-based visualization techniques can support decision-making by presenting infrastructure-related information in an accessible form. However, scalable visualization systems can be difficult to design and maintain [3].

Geospatial technologies have also been investigated for road condition reporting. GIS-based approaches can improve the identification and visualization of road problems, although their effectiveness depends on accurate geographic information [4].

Machine-vision techniques provide opportunities for automatic detection of road cracks and potholes. These methods can improve inspection efficiency, but they generally require appropriate datasets and computational resources [5], [7].

E-governance platforms can improve transparency and grievance redressal by providing digital channels through which citizens can communicate with public authorities [6]. However, adoption, accessibility, and scalability can influence the effectiveness of such systems.

Accessibility standards are important for public-sector applications because users may have different technical abilities and accessibility requirements [8]. Similarly, smart-city complaint systems can improve accountability but may introduce implementation and deployment complexities [9].

GIS-based monitoring frameworks have been proposed to support infrastructure maintenance activities. Such frameworks can provide improved spatial monitoring, although suitable geospatial infrastructure is required [10].

Mobile reporting combined with GIS can support road maintenance planning by connecting citizen reports with geographical information [11]. Complaint evaluation metrics can also be used to measure the responsiveness of complaint management systems, although universal evaluation standards are not always available [12].

Mobile civic reporting systems can improve citizen engagement, but usability may vary between different groups of users [13]. UI/UX design principles are therefore important for citizen-centric applications [14].

Responsive cross-platform interfaces provide compatibility across different devices. However, maintaining consistent behavior and design across platforms can increase development complexity [15].

Error prevention, error handling, and feedback mechanisms can improve system reliability and user satisfaction in public digital services [16]. Comparative studies of mobile complaint reporting tools also indicate that different solutions provide different advantages and limitations [17].

Complaint prioritization models can help ensure that critical service requests are addressed first. However, determining an appropriate priority level for every complaint can be difficult [18].

Weighted prioritization models can support maintenance scheduling and municipal decision-making. Their performance depends on the availability and reliability of input data [19].

Citizen participation models encourage public involvement in digital governance and civic issue reporting. Nevertheless, participation levels can differ among user groups [20].

Usability testing of government applications has been associated with improvements in efficiency and user satisfaction. Continuous testing and regular updates are important for maintaining application quality [21].

Service Level Agreement (SLA) estimation techniques can be used to estimate complaint-resolution time. However, resolution time can vary according to the type and complexity of the reported issue [22].

Probabilistic scoring methods can be used to improve service request prioritization. Such methods can improve decision-making but may introduce additional computational complexity [23].

Road maintenance is also connected with broader transportation safety and infrastructure-management challenges. International organizations have highlighted the importance of improving road safety and maintaining adequate infrastructure [24], [25].

The reviewed studies demonstrate that road complaint management can benefit from the combination of GIS, mobile/web reporting, image processing, prioritization, analytics, and e-governance technologies. However, integrating these capabilities into a single complaint-management workflow remains an important practical requirement.

III. RESEARCH GAP

The literature indicates that several technologies have been individually investigated for road maintenance and citizen complaint management. Image-processing approaches focus mainly on automated road-defect detection, while GIS-based systems focus on geographical visualization and monitoring [1], [4], [5], [10].

Similarly, mobile and web-based civic reporting systems provide digital complaint submission, but different systems may have limitations related to usability, accessibility, prioritization, and integration with maintenance workflows [13], [14], [17].

A significant requirement is therefore a centralized system that connects the complete complaint lifecycle. This lifecycle should include citizen registration, complaint submission, image upload, GPS-based location selection, administrator verification, task assignment, worker progress updates, completion evidence, citizen tracking, and analytical reporting.

The proposed RCMS addresses this gap by integrating these activities into a single platform. Instead of treating complaint submission, geographic information, maintenance activities, and reporting as separate processes, the proposed system connects them through a centralized database and role-based workflow.

The system also provides a foundation for future integration of AI-based road-damage detection and complaint-prioritization techniques. Such integration could reduce manual inspection effort and support more efficient maintenance planning [7], [18], [19].

IV. PROPOSED METHODOLOGY

The proposed Road Complaint Management System follows a structured workflow to simplify the reporting, verification, assignment, monitoring, and resolution of road-related complaints.

The major phases of the methodology are described below.

A. User Registration and Authentication

Citizens register in the system using the required personal information and securely log into the application. Authentication ensures that authorized users can submit and track complaints.

B. Complaint Registration

After successful authentication, citizens can submit a complaint by providing the following information:

- Complaint title.
- Description of the road issue.
- Image of the damaged road.
- GPS location.
- Complaint category.

The submitted information is stored in the centralized database for further processing.

C. Complaint Verification

The administrator reviews submitted complaints to verify their authenticity. Duplicate or invalid complaints can be rejected, while valid complaints are approved for further action.

This verification stage helps improve the reliability of the complaint database and reduces unnecessary duplication.

D. Task Assignment

After verification, complaints are assigned to an appropriate maintenance department or field worker. Assignment can consider the complaint location and category.

E. Repair Status Monitoring

Workers update the complaint status during different stages of the repair process. The proposed system uses the following basic status values:

- Pending
- In Progress
- Completed

Citizens can monitor the status of their complaints.

F. Complaint Resolution

After the repair is completed, the responsible worker uploads proof of completion. The administrator verifies the repair and the complaint is subsequently marked as resolved.

G. Data Management and Reporting

All complaint records are stored in the database. The system can generate reports according to complaint category, geographical location, resolution time, complaint status, and overall system performance.

These reports can support the identification of frequently affected areas and future road maintenance planning [10], [11].

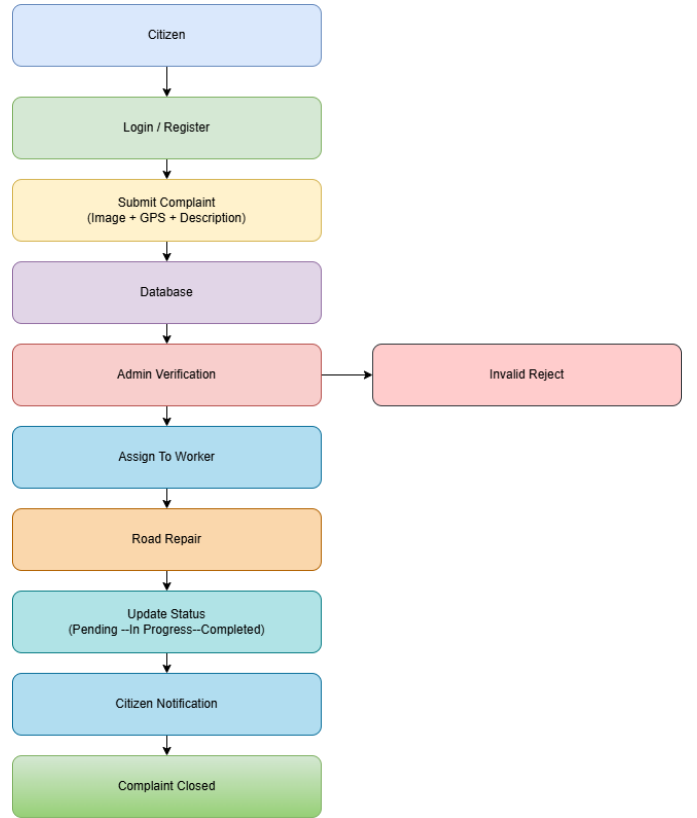


Fig. 1. Workflow of the Proposed Road Complaint Management System

V. SYSTEM ARCHITECTURE

The proposed Road Complaint Management System follows a client-server architecture that supports communication between citizens, administrators, maintenance workers, and the database.

The architecture consists of four major modules:

- 1) User Module
- 2) Administrator Module
- 3) Worker Module
- 4) Database Module

A. User Module

The User Module is designed for citizens. Users can:

- Register and log into the system.
- Submit road complaints.
- Upload road-damage images.
- Provide complaint descriptions.
- Select the complaint location using GPS or a map.
- Track complaint status.
- Receive status updates.

Location-based reporting provides authorities with geographical information about the reported road problem [4], [11].

B. Administrator Module

The Administrator Module provides centralized complaint management. Administrators can:

- View submitted complaints.
- Verify complaints.
- Reject duplicate or invalid complaints.
- Assign complaints to workers.
- Monitor repair progress.
- Manage users.
- Review completion evidence.
- Generate reports.

Role-based access can help ensure that administrative functions are available only to authorized users.

C. Worker Module

The Worker Module supports field maintenance staff. Workers can view complaints assigned to them, inspect the reported location, perform the required repair work, and update the repair status.

Workers can also upload an image or other evidence after completion of the repair.

D. Database Module

The Database Module stores information associated with:

- Users.
- Complaints.
- Complaint categories.
- Images.
- GPS coordinates.
- Complaint status.
- Worker assignments.
- Repair records.

Centralized storage allows complaint information to be retrieved and analyzed systematically.

VI. RESULTS AND DISCUSSION

The proposed Road Complaint Management System was developed to simplify the reporting and management of road-related issues. The system was evaluated with respect to complaint submission, usability, administrative management, repair tracking, and overall complaint-handling efficiency.

Citizens can register successfully, submit complaints with images and GPS locations, and monitor the status of submitted complaints.

The administrator dashboard provides a centralized interface for complaint verification and monitoring. Duplicate or invalid complaints can be identified and rejected, thereby improving the reliability of the complaint database.

Maintenance workers receive assigned complaints and can update their status as Pending, In Progress, or Completed. This provides a structured workflow for monitoring repair activities.

The proposed system improves communication between citizens and municipal authorities by providing a centralized digital complaint channel. The approach also improves transparency by allowing users to monitor the progress of their complaints.

The home page provides a simple interface through which citizens can access road complaint services. Navigation options can provide access to the home page, showcase, about/support information, and administrator login.

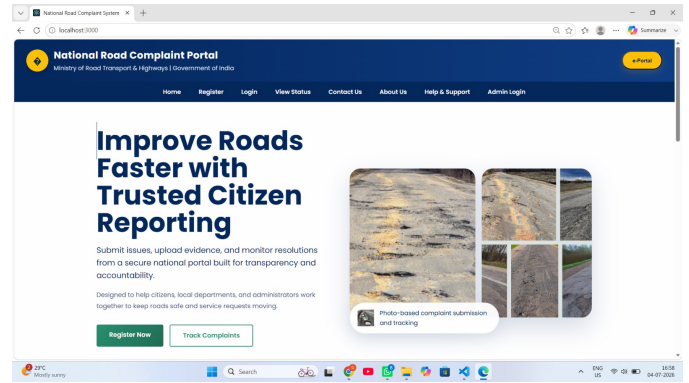


Fig. 2. Home Page of the National Road Complaint Portal

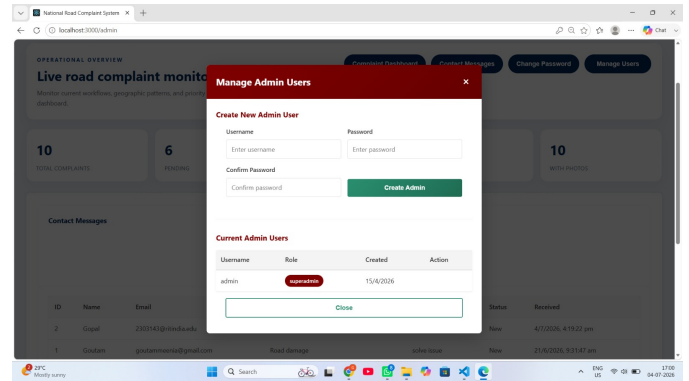


Fig. 3. Administrator User Management Interface

The administrator user-management interface provides controlled access to administrative functions. Authorized administrators can manage administrator accounts and their associated roles.

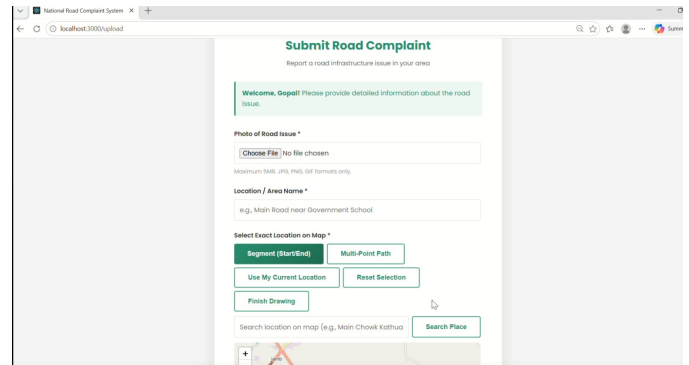


Fig. 4. Road Complaint Submission and Location Selection Interface

The complaint submission module enables citizens to provide details about road problems, upload images, and identify the affected location using a map. The location-selection facility helps associate the complaint with a geographical position.

The image-upload facility provides visual evidence of the reported road condition. Such evidence can support complaint

verification and maintenance prioritization.

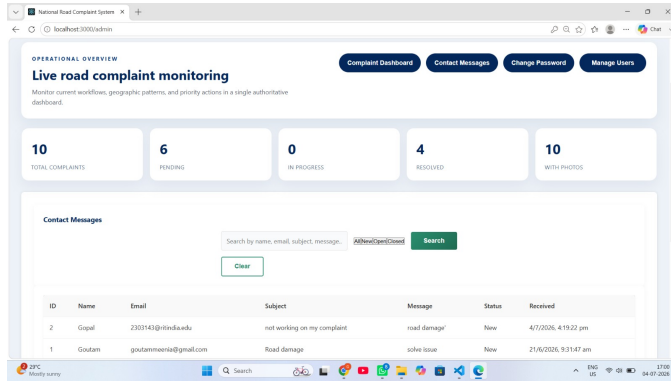


Fig. 5. Administrator Dashboard for Complaint Monitoring

The administrator dashboard provides a centralized overview of the complaint-management process. It can display complaint statistics such as total complaints, pending complaints, complaints in progress, resolved complaints, and complaints containing photographic evidence.

The dashboard can help administrators quickly understand the current complaint status and monitor communication received from citizens.

The results indicate that the proposed system provides a practical approach to improving the road complaint-management process. Traditional complaint channels can be supplemented by a centralized digital platform in which complaint information, photographs, and geographical information are collected systematically.

The system can further be enhanced by integrating an AI-based pothole and road-damage detection model. Such a model could automatically analyze uploaded road images and assist in identifying road defects [5], [7].

VII. CONCLUSION

The proposed Road Complaint Management System provides an intuitive and practical digital platform for reporting, monitoring, and managing road-related issues.

Citizens can submit complaints containing descriptions, images, and GPS-based locations. Municipal authorities can verify complaints, assign maintenance activities, monitor progress, and verify completion evidence.

The centralized complaint-management process reduces dependence on manual paperwork and provides improved communication between citizens and government authorities. The system supports complaint tracking and status updates, which can improve transparency and accountability.

The reviewed literature demonstrates the relevance of combining modern technologies such as image processing, computer vision, GIS, mobile/web applications, and digital public-service platforms for road maintenance and civic reporting [1], [4], [5], [10].

Future work can focus on integrating AI-based pothole detection, automatic complaint prioritization, advanced GIS visualization, predictive maintenance, and improved analytics.

These enhancements could support authorities in identifying high-risk areas and allocating maintenance resources more effectively.

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